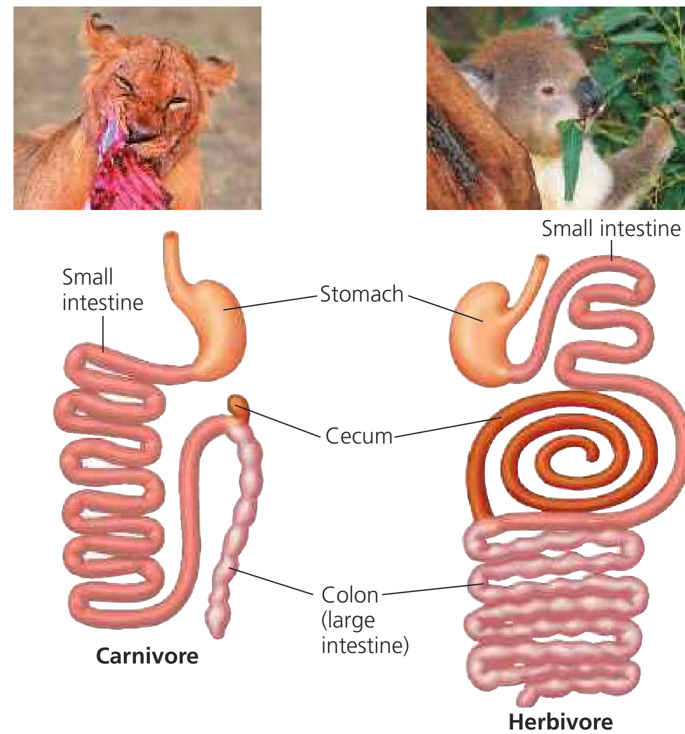
evolutionary adaptation of teeth for processing different kinds of food is one of the major reasons that mammals have been so successful. For example, a sea otter uses its sharp canine teeth to tear apart prey such as crabs and its slightly rounded molars to crush their shells. Nonmammalian vertebrates generally have less specialized dentition, but there are interesting exceptions. Venomous snakes, such as rattlesnakes, have fangs, modified teeth that inject venom into prey. Some fangs are hollow, like syringes, whereas others drip the toxin along grooves on the surfaces of the teeth.

Stomach and Intestinal Adaptations

Evolutionary adaptations to differences in diet are sometimes apparent as variations in the dimensions of digestive organs. For example, large, expandable stomachs are common in carnivorous vertebrates, which may wait a long time between meals and must eat as much as they can when they do catch prey. An expandable stomach enables a rock python to ingest a whole gazelle (see Figure 41.5) and a 200-kg African lion to consume 40 kg of meat in one meal!

Adaptation is also apparent in the length of the digestive system in different vertebrates. In general, herbivores and omnivores have longer alimentary canals relative to their body size than do carnivores. Plant matter is more difficult to digest than meat because it contains cell walls. A longer digestive tract furnishes more time for digestion and more surface area for nutrient absorption. As an example, consider the lion and koala in **Figure 41.16**. The koala has much longer intestines, relative to its size, enhancing the processing of fibrous, protein-poor eucalyptus leaves from which it obtains nearly all of its nutrients and water.

▼ Figure 41.16 The alimentary canals of a carnivore (lion) and herbivore (koala). The relatively short digestive tract of the lion is sufficient for digesting meat and absorbing its nutrients. In contrast, the koala's long alimentary canal is specialized for digesting eucalyptus leaves. Extensive chewing chops the leaves into tiny pieces, increasing exposure to digestive juices. In the long cecum and the upper portion of the colon, symbiotic bacteria further digest the shredded leaves, releasing nutrients that the koala can absorb.



Mutualistic Adaptations

An estimated 10–100 trillion bacteria live in the human digestive system. The coexistence of humans and many intestinal bacteria is an example of mutualism, an interaction between two species that benefits both of them (see Concept 54.1). For example, some intestinal bacteria produce vitamins, such as vitamin K, biotin, and folic acid, which are absorbed into the blood, supplementing our dietary intake. Intestinal bacteria also regulate the development of the intestinal epithelium and the function of the innate immune system. The bacteria in turn receive a steady supply of nutrients and a stable host environment.

Recently, we have greatly expanded our knowledge of the **microbiome**, the collection of microorganisms living in and on the body, along with their genetic material. To study the microbiome, scientists are using a DNA sequencing approach based on the polymerase chain reaction (PCR, see Figure 20.8). To date they have found more than 400 bacterial species in the human digestive tract, a far greater number than had been identified through approaches relying on laboratory culture and characterization. Furthermore, researchers have found significant differences in the microbiome associated with diet, disease, and age **(Figure 41.17)**.

One example of the power of microbiome studies comes from research on gastric ulcers, a disease that damages the stomach lining. Experiments demonstrating that ulcers are caused by infection with the acid-tolerant bacterium *Helicobacter pylori* and can be cured with antibiotics earned Australian researchers Barry Marshall and Robin Warren a Nobel Prize in 2005.

Recently, scientists analyzed the microbiome in samples from human stomachs to learn how *H. pylori* infection leads to ulcers. Their findings were dramatic: *H. pylori* infection led to the near-complete elimination of all bacterial species commonly found in the stomach **(Figure 41.18)**.

Studies on the microbiome have already led to therapies for intestinal infections by antibiotic-resistant pathogens, a major public health problem. One such therapy is fecal microbial transplantation, in which the microbiome of a healthy individual is introduced into the patient's intestine. This approach has been used for untreatable diarrhea caused